EXPERIMENT 5

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Experiment 5 – SQL Conditional Logic (Odd & Even Values)

Experiment 5: SQL Conditional Logic using the MOD (%) operator to classify employee salary values as Odd or Even. This experiment demonstrates numerical analysis and conditional evaluation using SQL queries in Oracle and PostgreSQL.

Aim
To understand and apply conditional logic in SQL by using the modulus operator (MOD / %) to analyze numerical data and classify employee salaries as odd or even, thereby improving data analysis and decision-making skills in SQL.

Objective
• To write and execute SQL queries using the MOD (%) operator.
• To retrieve salary details from an employee table.
• To classify salaries into Odd and Even categories.
• To display odd and even salary values separately.

Software Requirements

Database Management System:
Oracle Database Express Edition (Oracle XE)
PostgreSQL Database

Database Administration Tool / Client Tool:
Oracle SQL Developer (for Oracle XE)
pgAdmin (for PostgreSQL)

Practical / Experiment Steps

1. Create an employee table with columns:
   id
   name

department
salary

2. Insert sample employee salary records into the table.
3. Write a SQL query to retrieve employee salary details.
4. Use the MOD(salary, 2) function (Oracle) or salary % 2 (PostgreSQL) to check whether the salary is odd or even.
5. Use a CASE statement to classify salaries as:
   'Odd Salary'
   'Even Salary'
6. Execute separate queries to:
   Display only employees with Odd salaries.
   Display only employees with Even salaries.
7. Observe and record the output results.

Procedure

1. Open Oracle SQL Developer or pgAdmin.
2. Connect to the database.
3. Create the employee table using CREATE TABLE.
4. Insert sample records using INSERT INTO.
5. Execute SELECT queries with the MOD operator.
6. Apply CASE statement for conditional classification.
7. Display results and capture screenshots.

Input / Output Details

Input

Table Name: employee

Columns:
id (Integer)
name (Varchar)
department (Varchar)
salary (Integer)

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Logic Used:
MOD(salary, 2) or salary % 2
CASE statement

Step-wise Output

Step 1 – Create employee Table

```sql
CREATE TABLE employee (
    id INT PRIMARY KEY,
    name VARCHAR(50),
    department VARCHAR(50),
    salary INT
);
```

Data Output | Messages | Notifications

```
CREATE TABLE

Query returned successfully in 57 msec.
```

Step 2 – Insert Data into employee Table

```sql
-- Step 2 - Insert Data into employee Table
INSERT INTO employee VALUES (1, 'Rahul', 'HR', 25001);
INSERT INTO employee VALUES (2, 'Sneha', 'Finance', 32000);
INSERT INTO employee VALUES (3, 'Amit', 'IT', 45015);
INSERT INTO employee VALUES (4, 'Neha', 'Marketing', 28000);
INSERT INTO employee VALUES (5, 'Vikram', 'Sales', 39011);

-- Step 3 - Display All Employee Salary Records
```

Data Output | Messages | Notifications

```
INSERT 0 1

Query returned successfully in 68 msec.
```

Step 3 – Display Employee Salary Records

```sql
SELECT id, name, department, salary
FROM employee;
```

Data Output | Messages | Notifications

Showing rows: 1 to 5 | Page No: 1 of 1

	id [PK] integer	name character varying (50)	department character varying (50)	salary integer
1	1	Rahul	HR	25001
2	2	Sneha	Finance	32000
3	3	Amit	IT	45015
4	4	Neha	Marketing	28000
5	5	Vikram	Sales	39011

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Step 4 – Classify Salaries as Odd or Even

```sql
-- Step 4 - Classify Salaries as Odd or Even
SELECT
    name,
    salary,
    CASE
        WHEN salary % 2 = 0 THEN 'Even Salary'
        ELSE 'Odd Salary'
    END AS salary_type
FROM employee;
```

Data Output Messages Notifications

Showing rows: 1 to 5

	name character varying (50)	salary integer	salary_type text
1	Rahul	25001	Odd Salary
2	Sneha	32000	Even Salary
3	Amit	45015	Odd Salary
4	Neha	28000	Even Salary
5	Vikram	39011	Odd Salary

Step 5 – Display Only Odd Salaries

```sql
-- Step 5 - Display Only Odd Salaries
SELECT
    name,
    salary
FROM employee
WHERE salary % 2 <> 0;
```

Data Output Messages Notifications

Showing rows: 1 to 3

	name character varying (50)	salary integer
1	Rahul	25001
2	Amit	45015
3	Vikram	39011

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Step 6 – Display Only Even Salaries

```sql
-- Step 6 - Display Only Even Salaries
SELECT
    name,
    salary
FROM employee
WHERE salary % 2 = 0;
```

Data Output Messages Notifications

Showing rows: 1 to 2

	name character varying (50)	salary integer
1	Sneha	32000
2	Neha	28000

Successfully run. Total

Learning Outcome

After completing this experiment, the learner will be able to:

- Understand conditional logic in SQL queries.
- Use the MOD (%) operator for numerical analysis.
- Classify numeric values into odd and even categories.
- Write efficient SQL queries for payroll data analysis.
- Apply SQL logic in real-world enterprise environments.